

Bayesian Linear Regression

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1 Recap of Frequentist Inference

1.1 The Frequentist Viewpoint

The key idea in frequentist statistics is that the observed dataset is only one realization of a random experiment.

Imagine that nature generates data according to the linear model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_m X_{im} + \varepsilon_i,$$

where

$$\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

If we repeatedly collected new datasets from the same population and recomputed the least squares estimator each time, we would obtain different values of

$$\hat{\beta}.$$

Thus the least squares estimator should itself be viewed as a random variable.

The goal of frequentist inference is to understand the distribution of this random variable. Once we know its distribution, we can construct confidence intervals and perform hypothesis tests.

1.2 Distribution of the Least Squares Estimator.

Using matrix notation, write

$$y = X\beta + \varepsilon,$$

where

$$\varepsilon \sim N(0, \sigma^2 I_n).$$

The least squares estimator is

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

Substituting

$$y = X\beta + \varepsilon$$

gives

$$\begin{aligned} \hat{\beta} &= (X^T X)^{-1} X^T (X\beta + \varepsilon) \\ &= \beta + (X^T X)^{-1} X^T \varepsilon. \end{aligned}$$

Now we can go two ways. We can assume that the rows of X are also random or assume that X is fixed. The dependence of $\hat{\beta}$ on X is complicated. We will assume that the design matrix X is fixed, the only randomness comes from ε .

Since ε is multivariate normal, it follows that $\hat{\beta}$ is also multivariate normal:

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1}).$$

This formula is one of the most important results in linear regression. It tells us that the least squares estimator is centered at the true coefficient vector β , and it quantifies the variability of the estimator through the covariance matrix

$$\sigma^2 (X^T X)^{-1}.$$

Looking at the j -th component, we obtain

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 (X^T X)_{j+1,j+1}^{-1}\right),$$

where $(X^T X)_{j+1,j+1}^{-1}$ denotes the appropriate diagonal entry of $(X^T X)^{-1}$.

Consequently,

$$\frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{(X^T X)_{j+1,j+1}^{-1}}} \sim N(0, 1).$$

1.3 Estimating the Noise Level.

The previous result assumes that the noise standard deviation σ is known. In practice, σ is almost never known and must be estimated from the data.

The residual sum of squares is

$$RSS = \|y - X\hat{\beta}\|^2.$$

The standard estimate of σ is the residual standard error

$$\hat{\sigma} = \sqrt{\frac{RSS}{n - m - 1}}.$$

After replacing σ by $\hat{\sigma}$, the standard normal distribution is replaced by a Student t -distribution:

$$\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma} \sqrt{(X^T X)_{j+1,j+1}^{-1}}} \sim t_{n-m-1}.$$

Exercise: Justify the last two displays.

1.4 Confidence Intervals.

Let $t_{n-m-1,\alpha/2}$ denote the upper $\alpha/2$ -quantile of the t -distribution with $n-m-1$ degrees of freedom.

Then a $100(1 - \alpha)\%$ confidence interval for β_j is

$$\hat{\beta}_j \pm t_{n-m-1,\alpha/2} \hat{\sigma} \sqrt{(X^T X)_{j+1,j+1}^{-1}}.$$

The quantity

$$\hat{\sigma} \sqrt{(X^T X)_{j+1,j+1}^{-1}}$$

is called the standard error of $\hat{\beta}_j$.

A confidence interval is often misinterpreted as saying that β_j has probability $1 - \alpha$ of lying inside the interval. This is not the frequentist interpretation.

In the frequentist framework, the parameter β_j is fixed and unknown, while the interval is random because it depends on the random data. If we repeatedly generated new datasets and

constructed a confidence interval each time, then approximately $100(1 - \alpha)\%$ of those intervals would contain the true value of β_j .

2 Bayesian Linear Regression

The frequentist approach views the coefficient vector

$$\beta = (\beta_0, \beta_1, \dots, \beta_m)^T$$

as an unknown but fixed quantity.

The Bayesian approach takes a different point of view. Here the coefficients are treated as random variables. Before seeing the data, we express our uncertainty about the coefficients through a prior distribution. After observing the data, we update this prior using Bayes' rule and obtain a posterior distribution.

The posterior distribution contains all information about the unknown coefficients after observing the data.

We again assume the standard Gaussian linear model

$$y = X\beta + \varepsilon,$$

with

$$\varepsilon \sim N(0, \sigma^2 I_n).$$

2.1 A Default Prior.

A common default Bayesian choice is to place a very diffuse prior on all regression coefficients and on the logarithm of the noise level.

Formally, we may imagine

$$\beta_0, \beta_1, \dots, \beta_m, \log \sigma \stackrel{\text{ind}}{\sim} \text{Uniform}(-C, C),$$

where C is very large.

This prior expresses the idea that, before seeing the data, we have little information about the coefficients or the noise level.

2.2 Posterior Distribution

Under the Gaussian model, the likelihood is

$$p(y \mid \beta, \sigma) \propto \sigma^{-n} \exp \left(-\frac{1}{2\sigma^2} \|y - X\beta\|^2 \right).$$

The quantity

$$S(\beta) = \|y - X\beta\|^2$$

is the residual sum of squares.

Combining the likelihood with the diffuse prior gives the joint posterior density

$$p(\beta, \sigma \mid y) \propto \sigma^{-n} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) \cdot \frac{1}{\sigma},$$

where the extra factor $1/\sigma$ comes from using a flat prior on $\log \sigma$. Therefore

$$p(\beta, \sigma \mid y) \propto \sigma^{-n-1} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right).$$

To obtain the marginal posterior density of β , we integrate out σ :

$$p(\beta \mid y) \propto \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) d\sigma.$$

Let

$$u = \frac{\sqrt{S(\beta)}}{\sigma}.$$

Then

$$\sigma = \frac{\sqrt{S(\beta)}}{u}$$

and

$$d\sigma = -\sqrt{S(\beta)}u^{-2} du.$$

Thus

$$\begin{aligned} p(\beta \mid y) &\propto \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) d\sigma \\ &= \int_\infty^0 \left(\frac{\sqrt{S(\beta)}}{u}\right)^{-n-1} \exp\left(-\frac{u^2}{2}\right) \left(-\sqrt{S(\beta)}u^{-2}\right) du \\ &= (S(\beta))^{-n/2} \int_0^\infty u^{n-1} \exp\left(-\frac{u^2}{2}\right) du. \end{aligned}$$

The integral

$$\int_0^\infty u^{n-1} \exp\left(-\frac{u^2}{2}\right) du$$

does not depend on β . Therefore, as a function of β ,

$$p(\beta \mid y) \propto (S(\beta))^{-n/2}.$$

Remark 1. *The prior density is a decreasing function of $S(\beta)$ which makes sense. So the posterior mode is at the OLS estimate.*

Using the Pythagorean identity,

$$S(\beta) = S(\hat{\beta}) + (\beta - \hat{\beta})^T X^T X (\beta - \hat{\beta}).$$

Substituting this into the posterior density gives

$$p(\beta \mid y) \propto \left[\frac{1}{1 + (\beta - \hat{\beta})^T \frac{X^T X}{S(\hat{\beta})} (\beta - \hat{\beta})} \right]^{n/2}. \quad (1)$$

It turns out that this is exactly the density of a multivariate Student t -distribution.

2.3 The Multivariate Student t -Distribution

It is useful to briefly review the multivariate t distribution.

Definition. A random vector $T \in \mathbb{R}^p$ is said to have a multivariate t -distribution with location vector μ , scale matrix Σ , and ν degrees of freedom if its density is

$$f(t) = C \left(1 + \frac{1}{\nu} (t - \mu)^T \Sigma^{-1} (t - \mu) \right)^{-(\nu+p)/2},$$

where C is a normalizing constant chosen so that the density integrates to one.

We write

$$T \sim t_p(\mu, \Sigma, \nu).$$

The formula above should look familiar. Recall that if

$$X \sim N_p(\mu, \Sigma),$$

then the density of X is proportional to

$$\exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right).$$

Thus both the normal and t -distributions involve exactly the same quadratic form

$$(x - \mu)^T \Sigma^{-1} (x - \mu).$$

The difference is that the normal distribution has exponential tails, whereas the t -distribution has polynomial tails. Consequently, the t -distribution places more probability on extreme values.

The multivariate t -distribution can be obtained from a multivariate normal distribution.

Suppose

$$X \sim N_p(\mu, \Sigma)$$

and

$$V \sim \chi_\nu^2$$

are independent.

Define

$$T = \mu + \frac{X - \mu}{\sqrt{V/\nu}}.$$

Then

$$T \sim t_p(\mu, \Sigma, \nu).$$

One of the most useful properties of the multivariate t -distribution is that every linear combination of its coordinates is again t -distributed.

If

$$T \sim t_p(\mu, \Sigma, \nu)$$

and

$$a \in \mathbb{R}^p,$$

then

$$a^T T \sim t_1(a^T \mu, a^T \Sigma a, \nu).$$

In particular, each coordinate T_j satisfies

$$T_j \sim t_1(\mu_j, \Sigma_{jj}, \nu).$$

This fact will be extremely important later because it allows us to derive the posterior distribution of each regression coefficient β_j from the joint posterior distribution of the entire vector β .

2.3.1 Large Degrees of Freedom

When the degrees of freedom ν are large, the multivariate t -distribution becomes very close to a multivariate normal distribution.

To understand why, consider the density

$$f(t) \propto \left(1 + \frac{1}{\nu}(t - \mu)^T \Sigma^{-1}(t - \mu)\right)^{-(\nu+p)/2}.$$

When ν is large, the quantity

$$\frac{1}{\nu}(t - \mu)^T \Sigma^{-1}(t - \mu)$$

is small for values of t near the center of the distribution. Using the approximation

$$1 + z \approx e^z$$

for small z , we obtain

$$\left(1 + \frac{1}{\nu}(t - \mu)^T \Sigma^{-1}(t - \mu)\right)^{-(\nu+p)/2} \approx \exp\left(-\frac{\nu+p}{2\nu}(t - \mu)^T \Sigma^{-1}(t - \mu)\right).$$

Since

$$\frac{\nu+p}{\nu} = 1 + \frac{p}{\nu} \approx 1$$

when ν is large, the density becomes approximately

$$\exp\left(-\frac{1}{2}(t - \mu)^T \Sigma^{-1}(t - \mu)\right),$$

which is precisely the kernel of the multivariate normal density

$$N_p(\mu, \Sigma).$$

Another way to see this is through the representation

$$T = \mu + \frac{X - \mu}{\sqrt{V/\nu}},$$

where

$$X \sim N_p(\mu, \Sigma), \quad V \sim \chi_\nu^2.$$

By the Law of Large Numbers,

$$\frac{V}{\nu} \rightarrow 1$$

when ν becomes large. Thus

$$\sqrt{\frac{V}{\nu}} \approx 1,$$

and consequently

$$T \approx X.$$

Since X is multivariate normal, this again suggests that

$$t_p(\mu, \Sigma, \nu) \approx N_p(\mu, \Sigma) \quad (2)$$

for large ν .

2.4 Posterior Distribution of the Coefficients.

In light of the above definition of the multivariate t , by looking at (1), the posterior distribution is

$$\beta \mid y \sim t_{m+1} \left(\hat{\beta}, \frac{S(\hat{\beta})}{n-m-1} (X^T X)^{-1}, n-m-1 \right).$$

Define

$$\hat{\sigma} = \sqrt{\frac{S(\hat{\beta})}{n-m-1}}.$$

This quantity is exactly the residual standard error from classical linear regression.

With this notation,

$$\beta \mid y \sim t_{m+1} \left(\hat{\beta}, \hat{\sigma}^2 (X^T X)^{-1}, n-m-1 \right).$$

Thus the posterior is centered at the least squares estimate and its spread is determined by the same covariance structure that appears in frequentist inference.

An important property of the multivariate t -distribution is that every component is again t -distributed.

Therefore

$$\beta_j \mid y \sim t_1 \left(\hat{\beta}_j, \hat{\sigma}^2 (X^T X)^{-1}_{j+1,j+1}, n-m-1 \right).$$

Equivalently,

$$\frac{\beta_j - \hat{\beta}_j}{\hat{\sigma} \sqrt{(X^T X)^{-1}_{j+1,j+1}}} \mid y \sim t_{n-m-1}.$$

2.5 Bayesian Credible Intervals.

Let $t_{n-m-1, \alpha/2}$ denote the upper $\alpha/2$ -quantile of the t -distribution.

Then

$$P\left(\hat{\beta}_j - t_{n-m-1, \alpha/2} \hat{\sigma} \sqrt{(X^T X)_{j+1, j+1}^{-1}} \leq \beta_j \leq \hat{\beta}_j + t_{n-m-1, \alpha/2} \hat{\sigma} \sqrt{(X^T X)_{j+1, j+1}^{-1}} \mid y\right) = 1 - \alpha.$$

This interval is called a $100(1 - \alpha)\%$ Bayesian credible interval.

Notice the interpretation. After observing the data, the parameter β_j is treated as random. The credible interval therefore has the direct probabilistic interpretation

Given the observed data, there is probability $1 - \alpha$ that β_j lies in the interval.

This is fundamentally different from the frequentist interpretation of a confidence interval. No repeated draws from the model is necessary for this interpretation.

2.6 A Coincidence.

For the diffuse prior used above, the Bayesian credible interval turns out to be *numerically identical* to the classical frequentist confidence interval.

The calculations are different and the interpretations are different, but the final interval is exactly the same.

This coincidence is special to this particular model and prior choice. In general settings, Bayesian and frequentist intervals need not agree.

2.7 Large Sample Behavior.

The posterior distribution above is a multivariate t -distribution with $n - m - 1$ degrees of freedom.

When $n - m - 1$ is large, the t -distribution is very close to a normal distribution by (2). Therefore

$$\beta \mid y \approx N\left(\hat{\beta}, \hat{\sigma}^2 (X^T X)^{-1}\right).$$

Similarly,

$$\beta_j \mid y \approx N\left(\hat{\beta}_j, \hat{\sigma}^2 (X^T X)_{j+1, j+1}^{-1}\right).$$

Thus for large sample sizes, Bayesian uncertainty quantification for linear regression becomes approximately Gaussian.

3 Autoregression in Time Series

We now consider an example which is useful to illustrate the differences between frequentist and bayesian inference. Suppose we observe a time series

$$y_1, \dots, y_n.$$

The AR(1) model is

$$y_t = \beta_0 + \beta_1 y_{t-1} + \varepsilon_t, \quad t = 2, \dots, n.$$

Question: When would such a model be appropriate?

We will treat y_1 as deterministic. This is a linear model with response y_t and predictor y_{t-1} .

Define

$$y = \begin{pmatrix} y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix},$$

and

$$X = \begin{pmatrix} 1 & y_1 \\ 1 & y_2 \\ \vdots & \vdots \\ 1 & y_{n-1} \end{pmatrix}.$$

Then

$$y = X\beta + \varepsilon,$$

where

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

The least squares estimator still has the formal expression

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

However, here X contains

$$y_1, \dots, y_{n-1},$$

which are part of the observed time series. Therefore X is random and dependent on the response vector. In particular, we can no longer use the fact that $\hat{\beta}$ being a linear function of y is also normally distributed.

For example,

$$\hat{\beta}_1 = \frac{\sum_{t=2}^n (y_t - \bar{y}_{2:n})(y_{t-1} - \bar{y}_{1:n-1})}{\sum_{t=2}^n (y_{t-1} - \bar{y}_{1:n-1})^2},$$

where

$$\bar{y}_{2:n} = \frac{y_2 + \cdots + y_n}{n-1},$$

and

$$\bar{y}_{1:n-1} = \frac{y_1 + \cdots + y_{n-1}}{n-1}.$$

The exact finite-sample distribution of $\hat{\beta}_1$ is generally quite complicated because the covariates are themselves random and dependent on the response process. Nevertheless, asymptotic arguments can be used to study the behavior of $\hat{\beta}_1$ when the sample size becomes large.

Under suitable regularity conditions, one can show that

$$\sqrt{n}(\hat{\beta}_1 - \beta_1) \xrightarrow{d} N(0, \tau^2),$$

for an appropriate asymptotic variance τ^2 . The proof relies on a central limit theorem for dependent random variables, reflecting the fact that the observations in a time series are not independent; For details, see Shumway and Stoffer.

For Bayesian inference, we again place the prior

$$\beta_0, \beta_1, \log \sigma \propto 1.$$

By the chain rule,

$$p(y_1, \dots, y_n \mid \beta, \sigma) = p(y_1 \mid \beta, \sigma) \prod_{t=2}^n p(y_t \mid y_1, \dots, y_{t-1}, \beta, \sigma).$$

Assume

$$\varepsilon_t \sim N(0, \sigma^2),$$

and that ε_t is independent of the past.

Then

$$y_t \mid y_1, \dots, y_{t-1}, \beta, \sigma \sim N(\beta_0 + \beta_1 y_{t-1}, \sigma^2).$$

If $p(y_1 \mid \beta, \sigma)$ does not depend on β, σ , then

$$L(\beta, \sigma) \propto \sigma^{-(n-1)} \exp \left[-\frac{1}{2\sigma^2} \sum_{t=2}^n (y_t - \beta_0 - \beta_1 y_{t-1})^2 \right].$$

In matrix notation,

$$L(\beta, \sigma) \propto \sigma^{-(n-1)} \exp \left[-\frac{1}{2\sigma^2} \|y - X\beta\|^2 \right].$$

This has exactly the same form as ordinary linear regression, except that the sample size is $n-1$. Since the prior is also the same, we can use the exact same reasoning to derive the posterior distribution of β as a multivariate t .

Since there are two regression coefficients, the degrees of freedom are

$$(n - 1) - 2 = n - 3.$$

Therefore

$$\beta \mid y \sim t_2 \left(\hat{\beta}, \frac{S(\hat{\beta})}{n - 3} (X^\top X)^{-1}, n - 3 \right).$$

Equivalently, with

$$\hat{\sigma}^2 = \frac{S(\hat{\beta})}{n - 3},$$

we have

$$\beta \mid y \sim t_2 \left(\hat{\beta}, \hat{\sigma}^2 (X^\top X)^{-1}, n - 3 \right).$$

Thus Bayesian inference for the AR(1) model gives a posterior t -distribution by essentially the same calculation as ordinary linear regression, while frequentist inference requires a more delicate analysis of the sampling distribution.

Remark 2. *The above is an example where the X matrix is random. In that case, to derive the sampling distribution of $\hat{\beta}$ as a frequentist, one needs to take the randomness of both X, y into account. For a Bayesian, the fact that X and y is random does not really matter once the likelihood and the prior has been decided on.*

Unlike in the frequentist approach, the Bayesian machinery does not look at other hypothetical data sets or repeated draws from the model. For the Bayesian, the data which is observed is fixed, and the only thing that matters is the form of the prior and the likelihood as a function of the unknown parameters. This means we do not have to think about a hard probability question such as what is the sampling distribution of the least squares coefficients.